

6 Historical Counting Systems

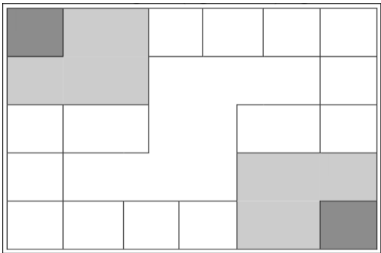
6.1 Tally Sticks

It is often conjectured that the earliest counting methods was 'tally sticks'.

This uses the idea of what we now call a one-to-one coorespondence. It is clear that this system is insufficient for most purposes.

6.1.1 The Inca Civilization

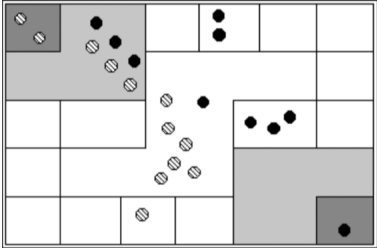
The Inca do not seem to have had a complicated counting system, which seems to be centered on record keeping. One way that they accomplished this was through the use of counting boards. Let's take a look.



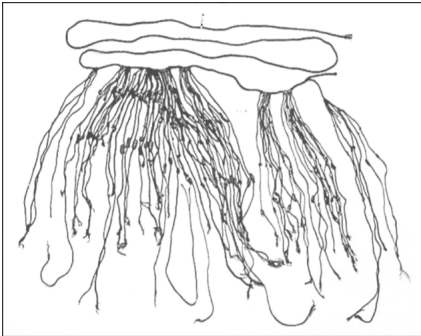
The outer twelve white boxes are ones
The two larger white boxes are twos
middle region is 3
light grey is 6
dark is 12

Let's assume that the black pebbles represent dogs, and the gray, cats. How many of each is found according to the following counting board.

41 dogs



The quipu was another method used for a more permanent form of counting, useful for keeping track of numbers over time.

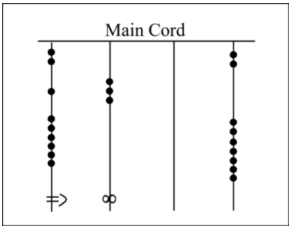


As we can see, these can get quite complicated, and are composed of strings (called H strings by some researchers) attached to a main cord. Counting was performed by tying a series of knots into the H strings according to the following:

- A figure 8 knot on the end of a string to denote 1
- A multiply looped knot to denote 2-9
- a single knot higher up on the string to denote 10, 100, 1000, or 10,000

Let's see an example.

Example 6.1.1 What numbers are represented on the following quipu.



The figure 8 denotes a figure 8 knot, the dots represent simple knots, and the lines with arc denote 2-9, depending on how many loops there are.

Example 6.1.2 Take your quipu, and use it to record the date of your birth, using one H string for the year, another for the month, and another for the day.

Note that this takes some forethought, placing the thousands too low will make it so in later strings you may not have enough room.

6.1.2 The Development of a System

The Brahmi numeral system was used in various locations throughout the world for several centuries, and consisted of symbols for many different numbers, which, when written, were added together to find the result. A table of some of the symbols is as follows:

	1	2	3	4	5	6	7	8	9
10 ⁰	—	=	≡	𑀓	𑀔	𑀕	𑀖	𑀗	𑀘
10 ¹	𑀠	𑀡	𑀢	𑀣	𑀤	𑀥	𑀦	𑀧	𑀨
10 ²	𑀩	𑀪	𑀫	𑀬	𑀭				
10 ³	𑀮	𑀯	𑀰	𑀱	𑀲				

Brahmi numerals signs of the 2nd century CE. 

No symbol for zero! So this cannot be a positional system like the one we use today.

Image taken from https://en.wikipedia.org/wiki/Brahmi_numerals

Is there anything weird (to us anyway) about this number system?

Let’s take a look at how our current system developed in the textbook.

6.2 Positional Systems

As we have noticed, the notion of ‘position’ is extremely valuable, but it took a long time for this idea to develop. Let’s discuss some of the basics of what a positional system is.

Write a number, explain briefly what each position is, this is pretty basic. Note that ancient civilizations did not necessarily use 10 as their base.